

A.2.5 Transient Input-Buffer Fault Sensitivity

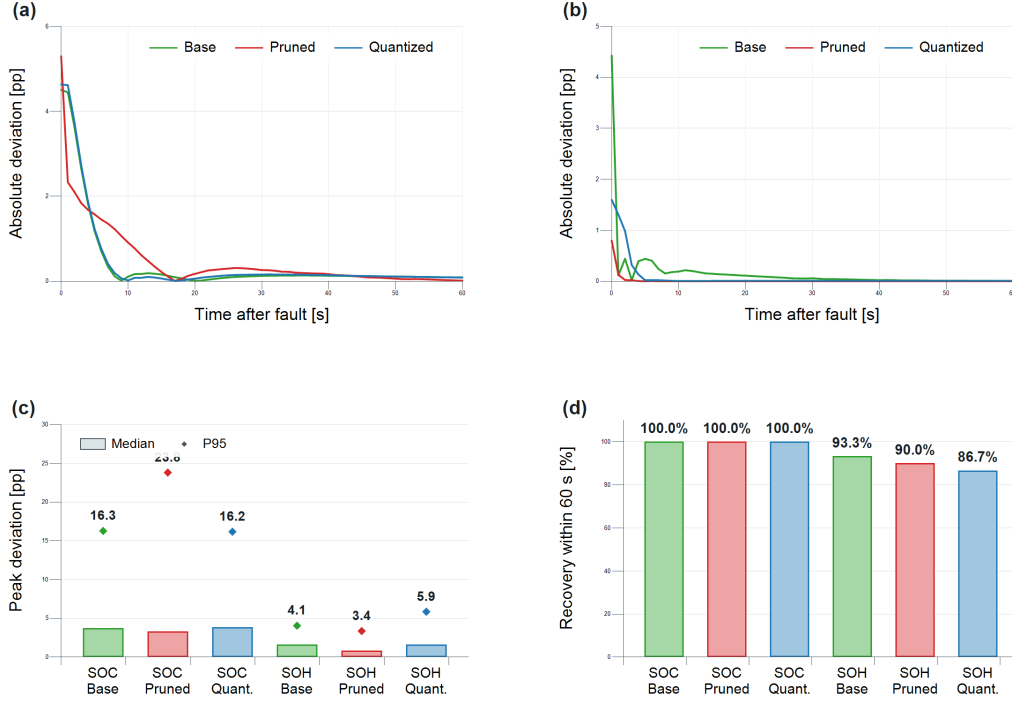


Figure A.8: Transient response of the clean and faulted model branches after a one-sample FP32 fraction-bit change in voltage, current, or temperature. The panels report peak disturbed-to-clean deviation, recovery within the following 60 s, and the target-error change over the complete 61-sample event window.

Table A.7: Median and P95 of the event-wise change in target MAE over each 61-sample fault window. All entries are percentage points.

Task	Model	Median ΔMAE_{61}	P95 ΔMAE_{61}
SOC	Base	0.0740	1.4235
SOC	Pruned	0.0464	1.5867
SOC	Quantized	0.0785	1.4314
SOH	Base	−0.0009	0.1224
SOH	Pruned	−0.00004	0.0533
SOH	Quantized	0.0012	0.1055